Vaibhav Tiwari

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EDUCATION

VIT Bhopal University *2023-Current*
Bachelor of Technology in Computer Science and Engineering with Specialization in AI/ML
Ragendra Swarup Public School, Agra
ISC (Class 12th, *2022*) Percentage: 97.0/100
ICSE (Class 10th, *2020*) Percentage: 92.4/100
Machine Learning Specialization by Stanford Online
Deep Learning Specialization by DeepLearning.Ai
Mathematical Methods for Quantitative Finance by MITx
Sequences, Time Series, and Prediction by DeepLearning.AI

EXPERIENCE

Summer Intern, FOSSEE IIT Bombay *February 2026 – Present*

- Selected for the Semester Long Internship under FOSSEE, IIT Bombay, to contribute to the Osdag open-source structural engineering project.
- Contributing clean, modular, and well-documented code to an open-source research codebase.

Independent Quantitative & Analytics ML Engineer *October 2024 – Present*

- Designed and implemented end-to-end machine learning pipelines for financial analytics, including sentiment-driven signals, time series forecasting, and systematic strategy research.
- Built modular workflows covering data ingestion, preprocessing, feature engineering, model training, and evaluation on noisy market and textual datasets.
- Evaluated models using statistical, error-based, and risk-aware metrics (rolling-window validation, robustness analysis) and translated insights into practical decision-support systems.

Sr.Team Member, **Music & Band Club-VIT Bhopal University** *April 2025 – February 2026*
Events Coordinator, Entrepreneurship Cell-VIT Bhopal University *September 2024 – June 2025*

PROJECTS

Financial Sentiment Analysis & Stock Market Price Movement Correlation System

- Built an end-to-end NLP-driven quantitative pipeline to extract market sentiment from financial news and social media using transformer-based models (BERT, FinBERT).
- Processed and aligned sentiment signals with equity price time series across multiple lag horizons (1-5 trading days) to study short-term market impact.
- Observed statistically meaningful lead–lag correlations between sentiment shifts and short-horizon returns across selected equities, validating sentiment as a weak predictive signal.
- Evaluated incremental predictive contribution of sentiment features over price-only baselines using regression-based analysis and rolling-window evaluation.
- Designed the system as a deployable web-based analytics pipeline to enable real-time sentiment ingestion and signal visualization.

Bitcoin Volatility Forecasting using Ensemble Machine Learning

- Built an end-to-end machine learning pipeline in Python to forecast Bitcoin price volatility using ensemble-based regression models, including Random Forest and XGBoost.
- Engineered lag-based, rolling-window, and statistical features from historical OHLCV data to capture temporal market dynamics.
- Achieved strong predictive performance with a Mean Absolute Error (MAE) of ~1,250 USD and an R^2 score of 0.92, and reduced prediction error variance by 18% through hyperparameter tuning and feature optimization.
- Conducted exploratory volatility analysis to derive market insights, identifying peak average prices of ~$56,030 in 2024 and maximum annual drawdowns of ~$30,100 in 2022.

Signal-Based Systematic Trading Strategy on Indian Equities

- Developed a systematic trading strategy on a universe of 10–15 liquid Indian equities (NIFTY 50 constituents) using daily OHLCV data.
- Engineered predictive trading signals using technical indicators including moving averages (5–50 day), RSI (14-day), and realized volatility computed over rolling windows.
- Implemented a modular backtesting framework with daily rebalancing, position sizing, and portfolio-level aggregation over multi-year historical periods.
- Evaluated performance using risk-adjusted metrics such as annualized Sharpe ratio and maximum drawdown, incorporating realistic transaction costs of 10–20 bps per trade and rolling-window robustness analysis.

SKILLS

Languages: Python, C/C++, Java, SQL, HTML
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, Hugging Face Transformers
Tools: Git, Docker, Jupyter Notebook, PyCharm, Google Colab, VS Studio, MS Excel, MS PowerPoint, Canva
Machine Learning & Quantitative Concepts: Supervised and Unsupervised Learning, Ensemble Methods, Feature Engineering Model Evaluation, Time Series Analysis, Volatility Modelling, Risk Metrices (Sharpe Ratio, Maximum Drawdown), NLP